

More Examples on Applications of Stacks: Expression Conversion and Evaluation

Expression conversion and evaluation are among the most important applications of stacks in Data Structures. Compilers, calculators, interpreters, and programming languages use stack-based algorithms to process mathematical expressions.

1. Types of Expressions

Expression Type	Example	Description
Infix	A+B*C	Operator is between operands
Prefix	+A*BC	Operator comes before operands
Postfix	ABC*+	Operator comes after operands

Example 1: Infix → Postfix

Expression

A+B*C

Step-by-Step

Symbol	Action	Stack	Output
A	Operand → Output		A
+	Push	+	A
B	Operand	+	AB
*	Higher precedence → Push	+ *	AB
C	Operand	+ *	ABC
End	Pop all		ABC*+

Final Postfix

ABC*+

Example 2: Infix → Postfix

Expression

$(A+B) * C$

Symbol	Stack	Output
(	(	
A	(	A
+	(+	A
B	(+	AB
)		AB+
*	*	AB+
C	*	AB+C
End		AB+C*

Final Answer

$AB+C*$

Example 3: Infix → Postfix

Expression

$A * (B+C)$

Input	Stack	Output
A		A
*	*	A
(	*(	A
B	*(	AB
+	*(+	AB

Input	Stack	Output
C	*(+	ABC
)		ABC+
End		ABC+*

Final Answer

ABC+*

Example 4: Infix → Postfix

Expression

A+B*C-D

Conversion

Input	Stack	Output
A		A
+	+	A
B	+	AB
*	+*	AB
C	+*	ABC
-	-	ABC*+
D	-	ABC*+D
End		ABC*+D-

Final Postfix

ABC*+D-

Example 5: Infix → Postfix

Expression

$(A+B) * (C-D)$

Conversion

Input	Stack	Output
(	(	
A	(	A
+	(+	A
B	(+	AB
)		AB+
*	*	AB+
(	*(	AB+
C	*(	AB+C
-	*(-	AB+C
D	*(-	AB+CD
)		AB+CD-
End		AB+CD-*

Final Answer

$AB+CD-*$

Example 6: Infix → Prefix

Expression

$A+B*C$

Method

1. Reverse Infix

C*B+A

2. Convert to Postfix

CB*A+

3. Reverse

+A*BC

Final Prefix

+A*BC

Example 7: Infix → Prefix

Expression

(A+B) * (C-D)

Final Prefix

*+AB-CD

Example 8: Prefix → Infix

Prefix

*+ABC

Stack Working

Read from **Right to Left**

Input	Stack
C	C
B	B C
A	A B C
+	(A+B) C

Input	Stack
*	((A+B)*C)

Final Infix

(A+B) *C

Example 9: Postfix → Infix

Postfix

AB+C*

Read from Left to Right.

Input	Stack
A	A
B	A B
+	(A+B)
C	(A+B) C
*	((A+B)*C)

Final Infix

(A+B) *C

Example 10: Postfix → Prefix

Postfix

AB+C*

Input	Stack
A	A
B	A B

Input	Stack
+	+AB
C	+AB C
*	*+ABC

Final Prefix

*+ABC

Postfix Evaluation Examples

Example 1

Expression

23+

Stack

Input	Stack
2	2
3	2 3
+	5

Answer

5

Example 2

Expression

23*54**+

Equivalent

$$(2 \times 3) + (5 \times 4)$$

Stack

Input	Stack
2	2
3	2 3
*	6
5	6 5
4	6 5 4
*	6 20
+	26

Answer

26

Example 3

Expression

82/53*+

Equivalent

$$(8 \div 2) + (5 \times 3)$$

Evaluation

Input	Stack
8	8
2	8 2
/	4
5	4 5
3	4 5 3

Input	Stack
*	4 15
+	19

Answer

19

Example 4

Expression

$52+83-*$

Equivalent

$(5+2) \times (8-3)$

Evaluation

Input	Stack
5	5
2	5 2
+	7
8	7 8
3	7 8 3
-	7 5
*	35

Answer

35

Example 5

Expression

56+72- /

Equivalent

(5+6) / (7-2)

Evaluation

Input	Stack
5	5
6	5 6
+	11
7	11 7
2	11 7 2
-	11 5
/	2

Answer = 2 (integer division)

Prefix Evaluation Examples

Prefix expressions are evaluated from **Right to Left**.

Example 1

Prefix

+23

Evaluation

2+3 = 5

Example 2

Prefix

*+234

Equivalent

(2+3) × 4

Evaluation

5×4 = 20

Example 3

Prefix

+*234

Equivalent

(2×3) +4

Evaluation

6+4=10

Comparison of Expression Types

Expression	Infix	Prefix	Postfix
1	A+B	+AB	AB+
2	A+B*C	+A*BC	ABC*+
3	(A+B)*C	*+ABC	AB+C*
4	A*(B+C)	*A+BC	ABC+*
5	(A+B)*(C-D)	*+AB-CD	AB+CD-*

Time Complexity

Operation	Complexity
Infix $\rightarrow$ Postfix	$O(n)$
Infix $\rightarrow$ Prefix	$O(n)$
Postfix $\rightarrow$ Infix	$O(n)$
Prefix $\rightarrow$ Infix	$O(n)$
Postfix $\rightarrow$ Prefix	$O(n)$
Postfix Evaluation	$O(n)$
Prefix Evaluation	$O(n)$

where **n** is the number of symbols (operands and operators) in the expression.

Practice Problems

Convert the following expressions:

1. $A+B-C$
2. $A * (B+C/D)$
3. $(A+B) / (C-D)$
4. $A+B * C-D/E$
5. $(A+B+C) * D$

Evaluate the following postfix expressions:

1. $34+$
2. $82/$
3. $53+62-*$
4. $72+43-*$
5. $56+83-*$

These examples cover the most common expression conversion and evaluation problems typically asked in BCA, B.Tech, and university Data Structures examinations.